

# Math

## 22 QUESTIONS

### DIRECTIONS

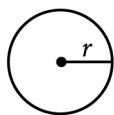
The questions in this section address a number of important math skills.  
Use of a calculator is permitted for all questions.

### NOTES

Unless otherwise indicated:

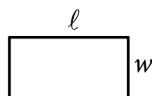
- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function  $f$  is the set of all real numbers  $x$  for which  $f(x)$  is a real number.

### REFERENCE

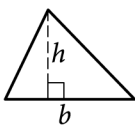


$$A = \pi r^2$$

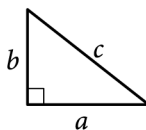
$$C = 2\pi r$$



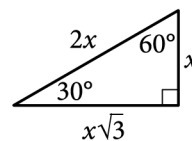
$$A = \ell w$$



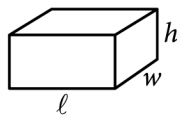
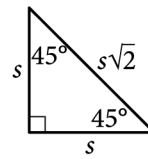
$$A = \frac{1}{2}bh$$



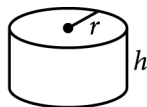
$$c^2 = a^2 + b^2$$



Special Right Triangles



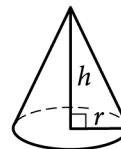
$$V = \ell wh$$



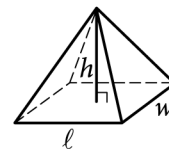
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is  $2\pi$ .

The sum of the measures in degrees of the angles of a triangle is 180.

**For multiple-choice questions**, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

**For student-produced response questions**, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as  $3\frac{1}{2}$ ), write it as an improper fraction ( $\frac{7}{2}$ ) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

Question ID: 7ce2830a

| Assessment | Test | Domain                            | Skill                                                                | Difficulty                                        |
|------------|------|-----------------------------------|----------------------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard <div><div></div><div></div><div></div></div> |

**Question**

A psychologist designed and conducted a study to determine whether playing a certain educational game increases middle school students’ accuracy in adding fractions. For the study, the psychologist chose a random sample of 35 students from all of the students at one of the middle schools in a large city. The psychologist found that students who played the game showed significant improvement in accuracy when adding fractions. What is the largest group to which the results of the study can be generalized?

- Answer**
- A. The 35 students in the sample
  - B. All students at the school
  - C. All middle school students in the city
  - D. All students in the city

Question ID: 0e709a29

| Assessment | Test | Domain                    | Skill                            | Difficulty                               |
|------------|------|---------------------------|----------------------------------|------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Right triangles and trigonometry | Hard <div></div> <div></div> <div></div> |

Question

$RS = 440$

$ST = 384$

$TR = 584$

The side lengths of right triangle  $RST$  are given. Triangle  $RST$  is similar to triangle  $UVW$ , where  $S$  corresponds to  $V$  and  $T$  corresponds to  $W$ . What is the value of  $\tan W$ ?

Answer

- A.  $\frac{48}{73}$
- B.  $\frac{55}{73}$
- C.  $\frac{48}{55}$
- D.  $\frac{55}{48}$

Question ID: f5ff91b2

| Assessment | Test | Domain  | Skill                            | Difficulty                                           |
|------------|------|---------|----------------------------------|------------------------------------------------------|
| SAT        | Math | Algebra | Linear equations in one variable | Hard<br><div><div></div><div></div><div></div></div> |

### Question

If  $\frac{x-5}{7} = \frac{x-5}{9}$ , the value of  $x - 5$  is between which of the following pairs of values?

**Answer**

- A. **-9** and **-7**
- B. **-3** and **3**
- C. **4.5** and **5.5**
- D. **6.75** and **9.25**

Question ID: d9137a84

| Assessment | Test | Domain        | Skill                  | Difficulty                                           |
|------------|------|---------------|------------------------|------------------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard<br><div><div></div><div></div><div></div></div> |

### Question

Which expression represents the product of  $(x^{-6}y^3z^5)$  and  $(x^4z^5 + y^8z^{-7})$ ?

**Answer**

- A.  $x^{-2}z^{10} + y^{11}z^{-2}$   
 B.  $x^{-2}z^{10} + x^{-6}z^{-2}$   
 C.  $x^{-2}y^3z^{10} + y^8z^{-7}$   
 D.  $x^{-2}y^3z^{10} + x^{-6}y^{11}z^{-2}$

## Question ID: 585de39a

| Assessment | Test | Domain                            | Skill                                   | Difficulty    |
|------------|------|-----------------------------------|-----------------------------------------|---------------|
| SAT        | Math | Problem-Solving and Data Analysis | Probability and conditional probability | Hard<br>■■■■■ |

**Question**

On May 10, 2015, there were 83 million Internet subscribers in Nigeria. The major Internet providers were MTN, Globacom, Airtel, Etisalat, and Visafone. By September 30, 2015, the number of Internet subscribers in Nigeria had increased to 97 million. If an Internet subscriber in Nigeria on September 30, 2015, is selected at random, the probability that the person selected was an MTN subscriber is 0.43. There were  $p$  million MTN subscribers in Nigeria on September 30, 2015. To the nearest integer, what is the value of  $p$  ?

**Question ID: 6ab30ce3**

| Assessment | Test | Domain                    | Skill                            | Difficulty                                           |
|------------|------|---------------------------|----------------------------------|------------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Right triangles and trigonometry | Hard<br><div><div></div><div></div><div></div></div> |

### Question

Triangle  $ABC$  is similar to triangle  $DEF$ , where  $A$  corresponds to  $D$  and  $C$  corresponds to  $F$ . Angles  $C$  and  $F$  are right angles. If  $\tan(A) = \sqrt{3}$  and  $DF = 125$ , what is the length of  $\overline{DE}$ ?

**Answer**

- A.  $125\frac{\sqrt{3}}{3}$   
B.  $125\frac{\sqrt{3}}{2}$   
C.  $125\sqrt{3}$   
D. 250



Question ID: 0366d965

| Assessment | Test | Domain  | Skill                             | Difficulty                               |
|------------|------|---------|-----------------------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | Hard <div></div> <div></div> <div></div> |

Question

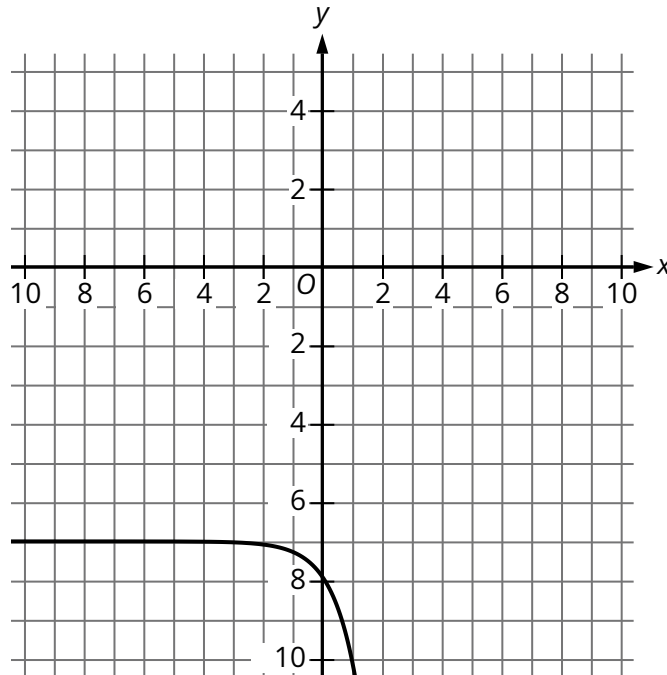
|    |    |
|----|----|
| x  | y  |
| 3  | 7  |
| k  | 11 |
| 12 | n  |

The table above shows the coordinates of three points on a line in the  $xy$ -plane, where  $k$  and  $n$  are constants. If the slope of the line is 2, what is the value of  $k + n$ ?

## Question ID: df71424b

| Assessment | Test | Domain        | Skill               | Difficulty    |
|------------|------|---------------|---------------------|---------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard<br>■■■■■ |

## Question



The graph of  $y = f(x)$  is shown, where  $f(x) = ab^x + c$ , and  $a$ ,  $b$ , and  $c$  are constants. For how many values of  $x$  does  $f(x) = 0$ ?

## Answer

- A. Three
- B. Two
- C. One
- D. Zero

## Question ID: 457d2f2c

| Assessment | Test | Domain                            | Skill                                                              | Difficulty    |
|------------|------|-----------------------------------|--------------------------------------------------------------------|---------------|
| SAT        | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard<br>■■■■■ |

**Question**

A data set of 27 different numbers has a mean of 33 and a median of 33. A new data set is created by adding 7 to each number in the original data set that is greater than the median and subtracting 7 from each number in the original data set that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

**Answer**

- A. Median
- B. Mean
- C. Sum of the numbers
- D. Standard deviation

Question ID: 7ce2e728

| Assessment | Test | Domain                    | Skill                        | Difficulty                                           |
|------------|------|---------------------------|------------------------------|------------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Lines, angles, and triangles | Hard<br><div><div></div><div></div><div></div></div> |

### Question

For triangles  $ABC$  and  $FGH$ , angles  $B$  and  $G$  each measure  $51^\circ$ , the length of  $\overline{AB}$  is 6, and the length of  $\overline{FG}$  is 18. Which additional piece(s) of information is(are) sufficient to prove that triangle  $ABC$  is similar to triangle  $FGH$ ?

- I. The measure of angle  $A$  is equal to the measure of angle  $F$ .
- II. The measure of angle  $C$  is equal to the measure of angle  $H$ .
- III. The length of  $\overline{FH}$  is 3 times the length of  $\overline{AC}$ .

**Answer**

- A. I is sufficient, II is sufficient, and III is sufficient.
- B. I is sufficient and II is sufficient, but III is not.
- C. II is sufficient and III is sufficient, but I is not.
- D. III is sufficient, but I is not and II is not.

## Question ID: d7bf55e1

| Assessment | Test | Domain  | Skill                                            | Difficulty    |
|------------|------|---------|--------------------------------------------------|---------------|
| SAT        | Math | Algebra | Systems of two linear equations in two variables | Hard<br>■■■■■ |

**Question**

A movie theater sells two types of tickets, adult tickets for \$12 and child tickets for \$8. If the theater sold 30 tickets for a total of \$300, how much, in dollars, was spent on adult tickets? (Disregard the \$ sign when gridding your answer.)

**Question ID: 91e7ea5e**

| Assessment | Test | Domain        | Skill               | Difficulty                                           |
|------------|------|---------------|---------------------|------------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard<br><div><div></div><div></div><div></div></div> |

### Question

$$h(x) = 2(x - 4)^2 - 32$$

The quadratic function  $h$  is defined as shown. In the  $xy$ -plane, the graph of  $y = h(x)$  intersects the  $x$ -axis at the points  $(0,0)$  and  $(t,0)$ , where  $t$  is a constant. What is the value of  $t$ ?

**Answer**

- A. 1  
B. 2  
C. 4  
D. 8

Question ID: 67c0200a

| Assessment | Test | Domain                            | Skill       | Difficulty                                        |
|------------|------|-----------------------------------|-------------|---------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Percentages | Hard <div><div></div><div></div><div></div></div> |

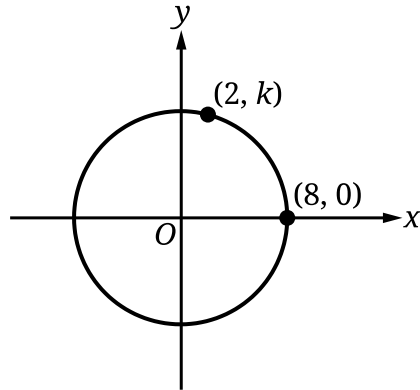
Question

The number  $a$  is 70% less than the positive number  $b$ . The number  $c$  is 80% greater than  $a$ . The number  $c$  is how many times  $b$ ?

## Question ID: d12bca0a

| Assessment | Test | Domain                    | Skill   | Difficulty    |
|------------|------|---------------------------|---------|---------------|
| SAT        | Math | Geometry and Trigonometry | Circles | Hard<br>■■■■■ |

## Question



Note: Figure not drawn to scale.

The circle shown has its center at  $(0, 0)$ . What is the value of  $k$ ?

## Answer

- A. 6
- B. 7
- C. 8
- D.  $\sqrt{60}$



| Assessment | Test | Domain  | Skill                             | Difficulty                                           |
|------------|------|---------|-----------------------------------|------------------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | Hard<br><div><div></div><div></div><div></div></div> |

| $x$   | $y$   |
|-------|-------|
| $-18$ | $-48$ |
| $7$   | $52$  |

**Answer**

- A.  $-\frac{4}{11}$   
B.  $-\frac{4}{77}$   
C.  $\frac{4}{7}$   
D.  $\frac{172}{7}$

Question ID: c95686e9

| Assessment | Test | Domain        | Skill               | Difficulty                               |
|------------|------|---------------|---------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div></div> <div></div> <div></div> |

Question

$$h(t) = -16t^2 + b$$

The function  $h$  estimates a ball's height, in feet, above a floor  $t$  seconds after the ball is released, where  $b$  is a constant. The function estimates that the ball is **19.36** feet above the floor when it is released at  $t = 0$ . How many seconds after being released does the function estimate the ball will reach the floor?

Question ID: ab176ad6

| Assessment | Test | Domain                    | Skill   | Difficulty                                        |
|------------|------|---------------------------|---------|---------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Circles | Hard <div><div></div><div></div><div></div></div> |

Question

The equation  $(x + 6)^2 + (y + 3)^2 = 121$  defines a circle in the  $xy$ -plane. What is the radius of the circle?

Question ID: 70feb725

| Assessment | Test | Domain  | Skill                                            | Difficulty                                        |
|------------|------|---------|--------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Systems of two linear equations in two variables | Hard <div><div></div><div></div><div></div></div> |

**Question**

During a month, Morgan ran  $r$  miles at 5 miles per hour and biked  $b$  miles at 10 miles per hour. She ran and biked a total of 200 miles that month, and she biked for twice as many hours as she ran. What is the total number of miles that Morgan biked during the month?

- Answer**
- A. 80
  - B. 100
  - C. 120
  - D. 160

Question ID: 9654add7

| Assessment | Test | Domain        | Skill               | Difficulty                                           |
|------------|------|---------------|---------------------|------------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard<br><div><div></div><div></div><div></div></div> |

### Question

$$f(x) = -500x^2 + 25,000x$$

The revenue  $f(x)$ , in dollars, that a company receives from sales of a product is given by the function  $f$  above, where  $x$  is the unit price, in dollars, of the product. The graph of  $y = f(x)$  in the  $xy$ -plane intersects the  $x$ -axis at 0 and  $a$ . What does  $a$  represent?

**Answer**

- A. The revenue, in dollars, when the unit price of the product is \$0
- B. The unit price, in dollars, of the product that will result in maximum revenue
- C. The unit price, in dollars, of the product that will result in a revenue of \$0
- D. The maximum revenue, in dollars, that the company can make

Question ID: 89661424

| Assessment | Test | Domain                    | Skill   | Difficulty                               |
|------------|------|---------------------------|---------|------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Circles | Hard <div></div> <div></div> <div></div> |

**Question**

A circle in the  $xy$ -plane has its center at  $(-5, 2)$  and has a radius of **9**. An equation of this circle is  $x^2 + y^2 + ax + by + c = 0$ , where  $a$ ,  $b$ , and  $c$  are constants. What is the value of  $c$ ?

**Question ID: 0b0fa68b**

| Assessment | Test | Domain  | Skill            | Difficulty                                           |
|------------|------|---------|------------------|------------------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Hard<br><div><div></div><div></div><div></div></div> |

### Question

For the function  $f$ ,  $f(cx) = x - 8$  for all values of  $x$ , where  $c$  is a positive constant. If  $f(2) = 35$ , what is the value of  $c$ ?

## Question ID: 7bd10ef3

| Assessment | Test | Domain        | Skill                                                                         | Difficulty    |
|------------|------|---------------|-------------------------------------------------------------------------------|---------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard<br>■■■■■ |

## Question

$$2x^2 - 4x = t$$

In the equation above,  $t$  is a constant. If the equation has no real solutions, which of the following could be the value of  $t$  ?

## Answer

A.  $-3$ B.  $-1$ 

C. 1

D. 3



## Math • Module 10 • Answers

Write one answer per line. SPR: integer, decimal, or fraction.

1. \_\_\_\_\_

2. \_\_\_\_\_

3. \_\_\_\_\_

4. \_\_\_\_\_

5. \_\_\_\_\_

6. \_\_\_\_\_

7. \_\_\_\_\_

8. \_\_\_\_\_

9. \_\_\_\_\_

10. \_\_\_\_\_

11. \_\_\_\_\_

12. \_\_\_\_\_

13. \_\_\_\_\_

14. \_\_\_\_\_

15. \_\_\_\_\_

16. \_\_\_\_\_

17. \_\_\_\_\_

18. \_\_\_\_\_

19. \_\_\_\_\_

20. \_\_\_\_\_

21. \_\_\_\_\_

22. \_\_\_\_\_